Interim Progress Report

A Survey of Deep Learning-Based Reconstruction Methods for Quantum Imaging Systems

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1. Introduction

This report covers the first three weeks of work on a literature survey of deep learning approaches to quantum image reconstruction. The project focuses on quantum ghost imaging, compressed sensing with entangled photons, sub-shot-noise imaging, and hybrid classical-quantum reconstruction methods.

2. Literature Review

A search across IEEE Xplore, arXiv, Google Scholar, and Optica Publishing using quantum imaging and deep learning keywords yielded around 45 relevant papers (1995–2025). These were classified by imaging modality, network architecture, and reconstruction task.

Ghost imaging has been covered in the most detail. The review traces the field from the original experiment by Pittman et al. (1995), through computational ghost imaging (Shapiro, 2008; Erkmen and Shapiro, 2010), to the first use of deep learning by Lyu et al. (2017), who reconstructed images using only 12.5% of the measurements needed by traditional correlation methods. The CNN approach of He et al. (2018), Shimobaba et al. (2018), and the end-to-end framework of Wang et al. (2019) with 3–5 dB PSNR improvement were also reviewed. Compressed sensing (Katz et al., 2009), sub-shot-noise imaging (Brida et al., 2010), and hybrid architectures (Beer et al., 2020; Killoran et al., 2019) have been partially covered.

3. Software Implementation

A Python/PyTorch codebase has been built and is available on GitHub. It contains:

Ghost imaging simulator (`src/simulator/`) – generates illumination patterns (Gaussian, binary, Hadamard), computes bucket detector signals with configurable noise, and performs correlation-based reconstruction.

Three reconstruction models (`src/models/`) – a fully connected network (Lyu et al., 2017), a CNN (He et al., 2018), and a U-Net (Ronneberger et al., 2015), all adapted for ghost imaging input.

Data pipeline and training (`src/data/`, `src/training/`) – MNIST-based dataset generation, training loop with early stopping, and evaluation with PSNR/SSIM metrics. The fully connected model at 12.5% sampling achieved 22 dB validation PSNR, matching published results.

4. Survey Manuscript

The survey document (`survey/`) is about 60–70% complete. Introduction, background, and ghost imaging sections are written (7 two-column pages). Compressed sensing, sub-shot-noise, hybrid methods, discussion, and conclusion are in draft. The bibliography has 22 entries. The draft manuscript is available in the repository.

5. Summary and Next Steps

Work done so far: systematic literature search, detailed review of ghost imaging methods, partial review of three other domains, full software framework with three working models, and a 60–70% complete survey manuscript. Remaining work: finish the outstanding reviews, run comparative experiments across all models and sampling ratios, and finalise the manuscript.

Key References: Pittman et al., *PRA* **52** (1995); Erkmen & Shapiro, *Adv. Opt. Photon.* **2** (2010); Lyu et al., *Sci. Rep.* **7** (2017); He et al., *Sci. Rep.* **8** (2018); Wang et al., *Opt. Exp.* **27** (2019); Katz et al., *APL* **95** (2009); Brida et al., *Nat. Photon.* **4** (2010); Barbastathis et al., *Optica* **6** (2019).